UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	PROGRAMMING I
COURSE CODE	:	CSC402
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC402 mock paper (November 2026). Generated answers show working; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 10 x 1 mark; Part B = 5 questions (60 marks); Part C = 2 x 15 marks; total = 100 marks.

4. Questions: 17. Slots: 22. Placeholders: 0. No LLM call. Questions and answers are for local study use only.

5. Structure derived from 21 unique CSC402 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A — ANSWER KEY

Q	Ans	Question	Solution
A1	A	Which phase of the program development life cycle defines what the program must do?	Analysis establishes the requirements before design and coding.
A2	B	What is displayed by the following code?	The loop displays 2, 4, 6 and 8 without spaces.
A3	C	What does the % operator return?	% is the modulus operator and returns the remainder.
A4	A	Which operator sends data to cout?	The insertion operator << writes to an output stream.
A5	A	Which of the following are valid identifiers?	total_score and _count are valid identifiers.
A6	C	Which statement immediately exits a loop?	break terminates the nearest loop or switch.
A7	B	Given the declaration, what is displayed?	word[3] is 't' because array indexing starts at zero.
A8	A	Which declaration creates an integer array of five elements?	int a[5]; reserves five integer elements.
A9	A	A computer system's hardware includes which of the following?	All five listed components are part of the hardware.
A10	A	Which part of a for loop is executed after each iteration?	The update expression runs after each iteration.

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PART B — WORKED ANSWERS

QUESTION 1

Topic: debugging testing

a) Define compiler and interpreter.

A compiler translates the whole source program before execution; an interpreter translates and executes one statement at a time.

Marking: Award 1 mark for each correct definition.

(2 marks)

b) State THREE differences between a compiler and an interpreter.

Compiler: whole program, produces object code, reports errors after compilation. Interpreter: statement by statement, no object file, stops at the first error; generally slower execution.

Marking: Award 1 mark for each valid difference.

(3 marks)

QUESTION 2

Topic: fundamentals

a) Draw a flowchart for the ticket price selection.

Start -> read age -> if age < 0 display invalid; else if age <= 12 price = 15; else if age <= 17 price = 25; else price = 40 -> display price -> end.

Marking: Award marks for validation, all ranges, output and flow.

(5 marks)

b) Write pseudocode and a complete C++ program for the ticket price selection.

Use if/else if with the three ranges and an invalid-age branch.

Marking: Award marks for validation, range conditions, assignment and output.

(8 marks)

QUESTION 3

Topic: data types

a) Write C++ declarations for: an integer age, a double price, a constant discount rate of 0.10 and a character grade.

int age; double price; const double DISCOUNT = 0.10; char grade;

Marking: Award marks for correct types, names and const initialisation.

(6 marks)

b) Write C++ statements to read age and price, calculate price after discount, and display the result.

cin >> age >> price; price = price - price * DISCOUNT; cout << price;

Marking: Award marks for input, discount expression and output.

(5 marks)

QUESTION 4

Topic: input output

a) Trace the program and state the output.

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Marking: The loop adds 1+2+3+4+5. Award marks for the sequence and final output.

(10 marks)

b) Rewrite the loop using a while loop. Then explain what changes if the condition is i < 5.

int i = 1; while (i <= 5) { total += i; i++; } With i < 5 the result is 10 because 5 is not added.

Marking: Award marks for correct while structure, update and changed result.

(5 marks)

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QUESTION 5

Topic: arrays

a) Declare two parallel arrays for the five sports in Table 1 and initialise them with those values.

string sports[5] = {"Swimming","Badminton","Athletics","Cycling","Hockey"}; int medals[5] = {8,12,7,5,9};

Marking: Award marks for both declarations, correct sizes and related indices.

(10 marks)

b) Write a program segment to display each sport with its medals, then calculate the total and average medals.

Loop i=0..4 to display sports[i] and medals[i], accumulate total, average = double(total)/5.

Marking: Award marks for paired display, total, average and output.

(6 marks)

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PART C — WORKED ANSWERS

QUESTION 1

A catering company charges the meal prices shown in Table 1.

Write a complete C++ program that reads the customer name and quantity, validates the quantity, calculates the total, displays a receipt, and repeats until the user chooses to stop. At the end, display the total quantity and total sales.

```
#include <iostream>
#include <iomanip>
#include <string>
using namespace std;
int main() {
    string name;
    int quantity, grandQuantity = 0;
    double unitPrice, total, grandTotal = 0;
    char again;
    do {
        cout << "Customer name: ";
        getline(cin >> ws, name);
        cout << "Quantity: ";
        cin >> quantity;
        if (quantity < 0) {
            cout << "Invalid quantity" << endl;
        } else {
            if (quantity >= 200) unitPrice = 9.50;
            else if (quantity >= 100) unitPrice = 10.50;
            else if (quantity >= 50) unitPrice = 11.50;
            else unitPrice = 12.50;
            total = quantity * unitPrice;
            grandTotal += total;
            grandQuantity += quantity;
            cout << fixed << setprecision(2);
            cout << "Receipt for " << name << ": "
                 << quantity << " x RM" << unitPrice
                 << " = RM" << total << endl;
        }
        cout << "Another customer? (y/n): ";
        cin >> again;
    } while (again == 'y' || again == 'Y');
    cout << "Total quantity: " << grandQuantity << endl;
    cout << "Total sales: RM" << fixed << setprecision(2)
         << grandTotal << endl;
    return 0;
}
```

Marking: Award 3 marks for input/validation, 6 for price selection and total, 3 for repetition, 2 for final totals and 1 for output clarity.

(15 marks)

QUESTION 2

A bookstore records up to five books. Table 1 shows an example of the fields.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

```
#include <iostream>
#include <string>
using namespace std;
struct Book {
    string title;
    double price;
    int quantity;
};
int main() {
    Book books[5];
    double totalValue = 0;
    int mostExpensive = 0;
```

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```
    for (int i = 0; i < 5; i++) {
        cin.ignore();
        cout << "Title: ";
        getline(cin, books[i].title);
        cout << "Price and quantity: ";
        cin >> books[i].price >> books[i].quantity;
        totalValue += books[i].price * books[i].quantity;
        if (books[i].price > books[mostExpensive].price) {
            mostExpensive = i;
        }
    }
    cout << "Total inventory value: RM" << totalValue << endl;
    cout << "Most expensive: " << books[mostExpensive].title
         << " (RM" << books[mostExpensive].price << ")" << endl;
    return 0;
}
```

Marking: Award 3 marks for struct definition, 4 for input, 4 for inventory value, 3 for most expensive and 1 for output.

(15 marks)

END OF MARKING SCHEME

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